

Assignment:- 5.

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Subject: Machine Learning.

① What is the purpose of the MME metric in regression?

→ The Mean Model (MME) is used to evaluate the average difference predicted and actual values in regression problems.

It tells how close the predictions are to the real outputs on average:

$$MME = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

A smaller MME means better performance. It's a simple and interpretable metric that, unlike Mean Squared Error, doesn't exaggerate large errors — making it more robust against outliers.

② Explain the concept of hyperparameter tuning in SVM.

→ In a SVM (Support Vector machine), hyperparameters determine how the model separates data. They aren't learned automatically but must be set manually or tuned.

Key hyperparameters include:

- C (Regularization): controls the trade-off between maximizing margin and minimizing misclassification.

- **kernel type:** Determines the function used to transform data (linear, polynomial, RBF, etc).
- **Gamma:** Influences the curvature of the decision boundary for non-linear kernels.

Hyperparameters tuning uses techniques like Grid search or Random search to test different combinations and finds the set that provides the best accuracy or F1-score on validation data.

③ What is the time complexity of predicting with a logistic regression model?

→ Prediction in logistic regression requires calculating a weighted sum of features (dot product) and applying the sigmoid function.

- For one sample $\rightarrow O(d)$
- For n samples $\rightarrow O(n \times d)$

It's fast because both the dot product and sigmoid are linear-time operations.

④ Define a soft-margin support vector machine

→ A soft margin SVM allows training points to be misclassified so the model can handle noisy or overlapping data.

It introduces slack variable (ϵ_i) and a penalty parameter (C).

$$\min_{w, b, \epsilon} \frac{1}{2} \|w\|^2 + C \sum_i \epsilon_i$$

A small C allows ^{more} margin violations (more generalization), while a large C enforces stricter classification with a narrower margin.

This flexibility helps in real-world data that's rarely perfectly separable.

⑤ What is the space complexity of a confusion matrix?

→ A confusion matrix for a classification task with k classes is a $k \times k$ table.

Each cell stores the count of predictions belonging to a particular actual-predicted combination.

Thus, the space complexity = $O(k^2)$.

For binary classification ($k=2$), it's fixed 2×2 matrix - constant space ($O(1)$).

⑥ Explain the concept of the AUC metric in classification.

→ AUC (Area under ~~ROC~~ ROC curves) measures how well a model distinguishes between positive and negative classes. It represents the probability that the model ranks a randomly chosen positive instance higher than a negative one.

- $AUC = 1$: perfect classifier
- $AUC = 0.5$: Random prediction
- $AUC < 0.5$: poor model

A higher AUC value means the model is better at separating classes across all threshold levels.

⑦ What is the time complexity of computing recall?

→ Recall measures the proportion of actual positives correctly identified:

$$\text{Recall} = \frac{TP}{TP + FN}$$

To compute recall, the algorithm scans all predictions once to count true positives (TP) and false negatives (FN).

Therefore, the time complexity = $O(n)$, where n is the number of samples.

⑧ What is the space complexity of a logistic regression model?

→ A logistic regression model stores:

- weights (w): one per feature $\rightarrow O(d)$
- Bias (b): single value $\rightarrow O(1)$

Here,

$$\text{total space complexity} = O(d)$$

This makes logistic regression memory-efficient and scalable to high-dimensional data.

⑨ Explain the concept of grid search in supervised learning.

→ Grid Search is a systematic hyperparameter optimization method used in supervised learning.

It builds a "grid" of parameter combinations, trains models for each, and selects the one with the best cross-validation score.

example:

for SVM:

params = {'C': [0, 1, 2, 10], 'kernel': ['linear', 'rbf']}

Each combination of C and kernel is tested - The one yielding the best accuracy or F1-score is chosen. While effective, Grid Search can be computationally heavy for large parameter spaces.

⑩ Define a multi-class classification task with an example.

→ A multi-class classification problem involves predicting one label from three or more classes.

Example:

Handwritten digit recognition (digit 0-9) using the MNIST dataset, each image belongs to one of ten classes.

Common algorithms include Decision Tree, Random Forests, Softmax Regression, and multi-class SVMs (one-vs-rest or one-vs-one approaches).

⑪ What is the time complexity of training a k-Nearest Neighbour model?

→ KNN doesn't have a true "training phase" it simply stores the dataset.

- Training time complexity : $O(1)$
- Prediction time complexity : $O(n \times d)$, since it computes distances between the test sample and all n training samples (each with d features).

This makes KNN slow during prediction but extremely fast to set up.

⑫ Define the precision-recall curve with an example

→ The precision-recall curve shows the how precision and recall vary as the decision threshold changes.

- Precision = $TP / (TP + FP)$
- Recall = $TP / (TP + FN)$

A good model has a curve close to the top-right corner (high precision and recall).

Example:

in fraud detection, lowering the threshold may increase recall (catch more frauds) but reduce precision (more false alarms).

(13) What is the space complexity of storing model predictions?

→ If a model produces predictions for n samples, and each prediction is a single scalar value, the space complexity is $O(n)$.

For multi-class problems with k classes (probabilities per class), the complexity becomes $O(n \times k)$.

(14) Explain the concept of a binary classification task.

→ Binary classification means predicting one of two possible categories.

Examples:

- Email spam detection (spam / not spam)
- Medical test result (Positive / Negative)
- Credit card fraud detection (Fraud / Normal)

It's the simplest classification type and evaluated using metrics like Accuracy, Precision, Recall, F1 and AUC.

like two classes (0 or 1)

Binomial

(15) What is the time complexity of computing the R^2 metric?

→ The R^2 (coefficient of determination) measures how well a regression model explains the variance in target

$$R^2 = 1 - \frac{SS_{res}(E)}{SS_{tot}(T)}$$

It requires computing predicted and actual differences once across all n samples, making the time complexity $= O(n)$.

It's efficient and widely used for regression model evaluation.

Long answer questions.

① Describe the applications of regression model in real-world scenarios.

→ Regression models are used to predict continuous values and understand relationships between variables.

Common application:

- Finance: Predicting stock prices, credit risk or loan defaults.
- Real Estate: Estimating house price based on features like area, location, and number of rooms.
- Healthcare: Forecasting disease progression or predicting patient recovery times.
- Marketing: Estimating sales or advertising impact using historical data.
- Environment: Predicting pollution levels or weather conditions.

② Explain the implementation of a soft-margin SVM with an example.

→ A soft-margin SVM allows some misclassified data points to handle noisy or overlapping datasets. It introduces slack variable (ξ) and a penalty parameter (C) to balance between maximizing the margin and minimizing misclassification.

Steps:

- 1) Normalize and prepare labeled data
- 2) Define the objective

$$\min_{w, b, \xi} \frac{1}{2} \|w\|^2 + C \sum \xi_i$$
- 3) Train using a kernel (like linear or RBF)
- 4) Predict new samples based on the decision boundary.

Example (Python):

```
from sklearn import svm
model = svm.SVC(kernel='linear', C=1.0)
model.fit(X_train, y_train)
```

Here,
 $C=1.0$ controls the trade off - smaller C gives a wider margin, larger C fits data more tightly.

③ Discuss the role of the AUC metric in evaluating classifier performance

- AUC (Area under Roc curve) measures how well a classifier separates classes. It represents the probability that a randomly chosen positive sample is ranked higher than a negative one.
- AUC = 1: perfect model
 - AUC = 0.5: Random guessing

It's especially useful for imbalanced datasets, as it evaluates performance across all thresholds rather than relying on a single accuracy value.

Example (Python):

```
from sklearn.metrics import roc_auc_score
auc = roc_auc_score(y_test, model.predict_proba(x_test)[:,1])
```

A higher AUC means better overall classification performance.

④ Explain the implementation of k-means clustering with a real-world example.

- k-means is an unsupervised learning algorithm used to group similar data point into k clusters.

Steps:

- 1) Choose the number of clusters (k)
- 2) Randomly initialize k centroids
- 3) Assign each data point to the nearest centroid.

- 4) update centroids as the mean of assigned points
- 5) Repeat until centroids stabilize

Example:

Customer segmentation - grouping customers by purchase behaviour to target marketing strategies.

Code:

```
from sklearn.cluster import KMeans
kmeans = KMeans (n_clusters=3, random_state=0)
kmeans.fit(x)
labels = kmeans.labels
```

⑤ What is the purpose of the MME metric in regression?

→ MME (Mean Model Error) measures the average absolute difference between predicted and actual values:

$$MME = \frac{1}{n} \sum |y_i - \hat{y}_i|$$

It gives a direct sense of prediction accuracy - lower MME indicates better model performance, compared to MSE, it's less sensitive to outliers and easier to interpret.

Example:

If an energy consumption model predicts 102 units when the actual usage is 100, the error is 2. Averaging such errors gives MME - a clear estimate of model accuracy.

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